

XIANGBO GAO (BARRY)

Software Developer Intern for Summer 2024 (Autonomous Driving and Perception)

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EDUCATION

MS in Robotics, University of Michigan Expected June 2025
BS in Computer Science & Mathematics, University of California, Irvine Sep. 2018 - Mar. 2023
Summer Session, University of California, Berkeley June 2019 - Sep. 2019
Related Courses: Advanced Computer Vision, Machine Learning, Deep Learning, Artificial Intelligence, Robotic Systems Laboratory, Mathematics for Robotics

SKILLS

Programming Language: Python, C/C++, Matlab
Software: PyTorch, ROS, OpenMMLab, Carla Simulator
Research/Development Backgrounds: ADAS, Perception, Deep Learning, Computer Vision, Motion Prediction

PUBLICATIONS

1. **X Gao**¹, et al. "Scale-free and Task-agnostic Attack: Generating Photo-realistic Adversarial Patterns with Patch Quilting Generator" arXiv:2208.06222, 2022 [Paper Link](#)
2. **X Gao**³, "Sample hardness based gradient loss for long-tailed cervical cell detection", International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 109-119. Springer, Cham, 2022 [Paper link](#)

INTERNSHIPS

Perception Algorithms Researcher Apr. 2023 - July 2023
🏢 Anhui Cowa ROBOT Co., Ltd. 📍 Shanghai, China
• Online HD Map Construction with Flow Map Prior, achieving 3.2% higher mAP than the baseline and 13th place in the CVPR Camera-based online HD map construction challenge 2023
• Motion Prediction with Historical Trajectories Clustering which solved over 95% of the existing corner cases.
Full-stack Software Developer June 2020 - Aug. 2020
🏢 Tandll Investment Management, Ltd 📍 Shenzhen, China
• Developed and deployed a management web using Django, MySQL, and React, which is currently in active use.
Software Developer Feb. 2019 - June 2019
🏢 Calit 2, University of California, Irvine 📍 Irvine, CA, USA
• Developed a virtual reality (VR) teaching tool for the MA6 Mask Aligner.

RESEARCH EXPERIENCE & LEADERSHIP

Institute of Computer Vision, Shenzhen University, China
Adversarial Attack with Semantic Pattern Apr. 2021 - Oct. 2022
• Proposed a novel PQ-GAN training strategy that learned a series of cascaded generators to manipulate image patterns at varying scales, demonstrating state-of-the-art attack strength and robustness against different defense algorithms.
Long-tailed Cervical Cell Detection May 2021 - Jan. 2022
• Proposed Grad-Libra Loss that uses gradients to dynamically calibrate sample hardness and rebalance gradients.
University of California, Irvine, CA
Auto-generated Graphical Model in the Autonomous Driving System May 2022 - Dec. 2022
• Designed and created a multi-domain ADAS dataset for various driving scenarios using the CARLA simulator.
• Developed a probabilistic LSTM structure that encodes states as variational embeddings using Pytorch.
Zero Waste Anteaters Sep. 2020 - Apr. 2022
• Led a team of eight to explore waste recognition solutions, achieving ~0.94 ACC for the image classification task and ~0.76 mAP for the object detection task.
• Designed an introductory Image Recognition tutorial and side projects for students interested in the field.

COMPETITIONS

Rank 13 th	CVPR Camera-based online HD map construction challenge 2023	May 2022
Outstanding Award	Netease Hackathon Competition	June 2020
Rank 1 st	Machine Learning Hackathon, University of California, Irvine, CA	April 2020
Rank 2 nd	Google Hash Code 2020 Algorithms Competition, Irvine, CA	Feb. 2020

LANGUAGES

Chinese (Native), English (Proficient)